# 2026 HCI Prelim H2 Geography Paper 1 ANSWERS

**1** Resource 1 shows the value of greenfield Chinese manufacturing FDI in ASEAN, by sector. Resource 2 shows export transition for Vietnam and China 2015 – 2024. Resource 3 shows flows of trade between China and Vietnam and Vietnam's GDP per capita by administrative divisions. Resource 4 shows an extract from an article on Vietnam's position in the global economy.

(a) Cite data from Resource 1 to describe Chinese manufacturing FDI in ASEAN in 2014 and 2024. [4]

Possible content:

- The total value of Chinese manufacturing FDI in ASEAN is much lower in 2014 at 0.7 USD billion than in 2024 at 9.3 USD billion. [1]
- In 2014, there were only two main components of FDI in automotives and consumer goods. [1]
- In 2024 Chinese manufacturing FDI in ASEAN was diversified into the renewable energy equipment and other sectors which were not included in 2014. [1]
- The largest increase in FDI between 2014 and 2024 was in ICT and electronic equipment which forms around 2 billion USD in 2024. [1]

(b) Account for Chinese manufacturing FDI in ASEAN shown in Resource 1. [6]

Possible content:

- Comparative advantage in factors of production such as lower land and labour costs, reduce production costs and increase profits [1]
- Higher and improving levels of development in places such as Vietnam provides established infrastructure to support manufacturing such as transport networks and reliable supply of water and energy.[1]
- Development of free trade zones and special economic zones reduce import and export tariffs which make it attractive to investors [1]
- Producing in host countries provide firms access to the consumer market by sidestepping import tariffs [1]
- Chinese firms avoid US tariffs by producing in Vietnam and exporting the goods to the US from Vietnam [1]
- Chinese firms may also use raw materials extracted in ASEAN countries by including them in the production process. [1] This reduces the cost of transport and hence the cost of production. [+1]

(c) Compare the value of exports for China and Vietnam shown in Resource 2. [6]

Possible content:
Similarities:

- Both Vietnam's and China's exports to US reflect very similar patterns with a sharp increase between 2015 – 2022, followed by a fall and then an increase to 2024. [1]

Differences:

- China's exports to US remains relatively high from 2015 – 2024 while Vietnam's exports to US experiences a significant Increase from 2015 to 2024. [1] The highest value half exports from China to the US was around 550 billion U.S. dollars in 2018, while the highest export value to Vietnam two US was around 140 billion U.S. dollars in 2024. [+1]
- While the export value of China to us this is fluctuating and experiences rises and falls over the years, Vietnam's export values to the US experiences a steady increase of around 100 billion U.S. dollars. [1]
- Furthermore, there has been a rising amount of exports from China to Vietnam from 2015 to 2014 from 50 billion U.S. dollars to 150 billion U.S. dollars, a 30% increase. [1]
- In 2018 China's exports peaked to a high value of around 550 billion U.S. dollars, followed by a significant decrease the year after. [1] The following year Vietnam experienced a significant increase and this has been rising ever since. [+1]

(d) With reference to Resource 3, suggest how flows of trade and capital may influence the level of development in Vietnam. [4]

Possible content:

- The flow of goods through Pingxiang and Dong Dang connect supply chains in China with factories in northern Vietnam which leads to the provision of greater job opportunities. [1] Such flows generate steady employment across logistics, customs clearance, freight forwarding, cold-chain warehousing, transport, and commercial service sectors. [+1]
- Such flows also encourage the development of efficient and rapid overland freight and other transport networks, [1] such that multinational electronics and automotive firms in Vietnam can operate lean, flexible supply chains without holding massive domestic inventories, increasing efficiency and growth of production plants. [+1]
- Capital flows can facilitate joint ventures and other forms of inter-firm networks in processing industries [1], enabling Vietnamese firms to capture a larger share of pre-export processing before goods cross into China, leading to the growth of production plants. [+1]
- Established trade routes may also encourage fresh produce from Vietnam such as tropical fruits and seafood from southern and central Vietnam to move through this channel to reach Chinese consumer markets. [1]
- The level of development across Vietnam can also be uneven as the areas and cities that receive investments tend to be more highly developed with higher GDP per capita, as shown by R3 [1] Areas where the flow of trade and capital is limited, the GDP per capita is lower as they do have access to the possibly thousands of direct and indirect jobs created in freight handling, customs brokering, warehousing, trucking, hospitality, and retail. [+1]

(e) With reference to Resources 3 and 4, explain why Vietnam's structure of the economy may change.[6]

Possible content:

- With improvements in transport and building infrastructure, there will be an increase in investments in the manufacturing industry. [1] Such increases are also supported by how factories from China choose to locate in open markets such as Vietnam to avoid US tariffs. [+1]
- The manufacturing sector is growing also because of the 'low-cost but also relatively skilled' labour in Vietnam. [1] Such people can be easily trained to take on work in production lines. [+1]
- With the high placing in the Human Capital Index, second only to Singapore in ASEAN, Vietnam can establish its position in high-tech manufacturing or higher order products such as semi-conductors as well as in the service industry such as R&D. [1] As the population possesses a higher level of productivity in the primary and secondary sectors, the real cost of essential goods (such as food, clothing, and basic appliances) drops significantly and real disposable income increases, leading to a higher demand in services such as healthcare, higher education, travel, entertainment, personal wellness, and financial planning. [+1]

(f) Suggest how Vietnam's economy and environment may be impacted as a host country in the global economy. [4]

Possible content:

Economic impact:

- There can be FDI-driven growth and export expansion. As a central hub for the "China + 1" strategy, Vietnam attracts significant Foreign Direct Investment (FDI) in electronics, machinery, textiles, and automotive components. [1] Industrial exports account for nearly 90% of total merchandise sales, driving steady national GDP growth. [+1]
- Vietnam can move up the value-chain. Increased capital inflows accelerate Vietnam's transition from basic, labour-intensive assembly toward higher-tech operations, [1] producing higher value-added products, including semiconductor packaging, renewable energy equipment, and consumer electronics manufacturing. [+1]

- There can be increased employment and household income growth. Expanding industrial processing zones generate millions of formal jobs in manufacturing, logistics, and corporate services, [1] accelerating urban transition and boosting domestic consumer purchasing power. [+1]

Environmental impact:

- Rapid industrial expansion places intense demand on the national electricity grid. [1] Short-term power generation reliance on fossil fuels risks increasing industrial carbon emissions and air pollution. [+1]
- High-density manufacturing corridors generate significant volumes of industrial wastewater, chemical runoff, air pollution, and electronic waste, [1] straining municipal waste management and local ecosystems especially in less developed areas like Vietnam. [+1]
- Export regulations in key foreign markets (such as the EU's Carbon Border Adjustment Mechanism) push domestic factories and eco-industrial parks to adopt environmentally friendly practices in the manufacturing process [1] such as wastewater recycling, energy-efficient production lines, and sustainable waste management practices. [+1]

**Cluster 2: Tropical Environments**

**2** Resource 5 shows the climate graph of Bujumbura, Burundi in east-central Africa. Resource 6 shows a mass movement in Bujumbura, Burundi. Resource 7 shows a photograph of a type of mass movement in the Valley of the Kings, Egypt. Resource 8 shows the influence of weathering on intact rock and rock mass properties.

(a) Cite data from Resource 5 to describe the characteristics of Bujumbura's climate. [4]

Point marked

- Low total annual rainfall, with figures [1]
- Seasonal rainfall, with identification of wet season and dry season [1]
- High mean annual temperature with figures [1]
- Small annual temperature range, with figures [1]

(b) With reference to Resources 5 and 6, explain the role of natural factors in contributing to the occurrence of mass movements.

*Point marked; any two-three natural factors, with further elaboration*

- Climate plays a dominant role in the occurrence of mass movements in Burundi by acting as both a short-term trigger and a long-term preconditioning factor. [1]
- As a region characterized by a humid tropical climate, Burundi experiences intense seasonal **rainfall** that facilitates rapid chemical weathering [1], both of which compromise slope stability by increasing shear stress/ decreasing shear strength [1]; particularly for shallow landslides such as debris slides and avalanches [1]
- During heavy rainstorms, water infiltrates slope materials, filling pore spaces and increasing pore-water pressure. This pressure pushes soil particles apart, acting as a lubricant that reduces the internal friction and shear strength of the material [1].
- **High mean annual temperatures** speed up the rate of chemical weathering and with abundant moisture accelerate the decomposition of minerals in the bedrock, leading to the development of thick regolith and saprolite layers. [1] This weathered material is significantly softer and has lower shear strength than fresh rock [1]
- **Other natural factors include: vegetation, slope angle and geology**

[6]

(c) With reference to Resources 6 and 7, compare the characteristics of the mass movements shown in Bujumbura and the Valley of the Kings. [4]

*Point marked; any four with at least one similarity and one difference*

Similarities

- In both contexts, weathering weakens the rock masses, transforming hard rock into less stable, transportable material [1]
- The occurrence of mass movement follows the same physical principle for both, taking place when the driving force of gravity (shear stress) exceeds the resisting force (shear strength**)** of the slope material [1]
- Speed of both landslides and rock falls are rapid [1]

Differences

- Dominant types of mass movement: Bujumbura is dominated by flow and slide processes [1]; in the Valley of the Kings, the dominant types are falls and block slides [1]
- Products of mass movement: The humid climate of Bujumbura facilitates deep chemical weathering, transforming hard bedrock into thick layers of soft regolith and residual soil.[1] In contrast, the resulting products in Valley of the Kings are mechanical fragments, such as angular blocks and talus, which retain the mineral composition of the parent rock [1]

(d) Suggest how human factors may have contributed to the occurrence and nature of the mass movement shown in Resource 7. [4]

*Point marked; any two factors with further elaboration*

- In the arid tropics where geomorphic processes are naturally dominated by physical weathering and moisture is scarce, human factors act as significant catalysts for mass movements by introducing moisture, chemicals, and mechanical instability [1]
- Mechanical weakening via engineering and excavation: Human construction activities, such as road building or mining, often employ tools and blasting techniques that cause new fractures to develop [1] or widen existing discontinuities in the rock mass [1]
- These fractures allow the rare instances of rainfall and air to infiltrate much deeper into the ground than would occur naturally, accelerating the breakdown of slope materials [1]
- The creation of cut slopes for infrastructure removes the natural "toe" or support of a slope, increasing the shear stress on the remaining material and making it susceptible to falls or slides [1]
- Leaking sewage, mining activities that release acids, and the use of fertilizers introduce water and chemicals into the subsurface to weaken the rock masses [1]
- Infrastructure in arid regions can concentrate runoff into road gutters and drainage zones [1]. During rare but intense rain events, this concentrated runoff can lead to localized saturation of slopes, triggering debris slides or flows [1] that would not have occurred under natural drainage conditions

(e) Explain the dominance of physical weathering processes over chemical weathering processes in arid environments. [6]

*Point marked; any two reasons with further elaboration*

- In arid environments, the extremely low and seasonal rainfall provides insufficient moisture to facilitate chemical reactions. [1]
- Without a consistent flow of water across or through rock surfaces, chemical weathering processes reliant on water e.g. carbonation and hydrolysis are severely inhibited [1], allowing physical disintegration to be the primary geomorphic process
- The radiative surplus and high temperatures of the arid tropics promote mechanical stress [1]
- Significant diurnal (day-to-night) temperature ranges cause rocks to repeatedly expand and contract and this mechanical stress/ thermal expansion leads to physical disintegration of the rock surface. [1]

- High rates of evaporation cause moisture to be drawn to the surface, leaving behind salt crystals in rock pores [1] and as these crystals grow, they exert physical pressure that pries the rock apart [+1]

(f) With reference to Resource 8 explain how rock weathering affects slope stability. [6]

*Point marked*

- R8 - Discontinuities drastically reduce the overall shear strength of a rock mass by creating planes of weakness where the rock can slide easily. Closer spacing results in more blocks, lower mass strength, and higher permeability.
- Weathering decreases the strength of the intact rock [1] and increases the spacing and nature of discontinuities by developing/ increasing fractures within the rock mass that act as planes of weakness. [1]
- As minerals are chemically altered or physically disintegrated, the tightness of the ground structure—the way particles and blocks are locked together—is reduced [1]
- Weathering agents like percolating water and air can then act on the “discontinuities” [1]. As weathering progresses, discontinuities become more persistent (extending further through the mass) and wider.[1] This increased aperture allows more water and air to circulate, creating a positive feedback loop that accelerates weathering deeper into the ground [1]
- In the humid tropics, chemical weathering transforms original hard minerals into softer clay minerals, creating a thick, weak layer of regolith that is highly susceptible to movement [1]
- Impact on slope stability – reducing shear strength of the discontinuity and altering the geo-mechanical characteristics of a rock mass, transforming hard rock into soft rock [1]

**3** 'Extractive industries can never truly contribute to the goals of sustainable development because they are based on the exploitation of non-renewable resources.'

Evaluate this statement. **[20]**

*Levels marked using the H2 generic levels descriptors for essays.*

This question requires students to synthesise the concept of sustainable development (meeting present and future needs while maximising economic, environmental, and social goals) with the characteristics of extractive industries, specifically the extraction of non-renewable and locationally specific resources.

***Possible approach***

Students can assess the trade-offs between the economic gains (development) and the long-term environmental and social impacts. They can also evaluate whether technological innovations or better management strategies can mitigate these impacts enough to align with Sustainable Development Goals (SDGs)

Some key points:

- Extractive industries contribute to sustainable development by funding local infrastructure, supplying materials for clean technologies, minimizing environmental footprints, and driving socio-economic capacity.
- Aligning mining, oil, and gas operations with the UN Sustainable Development Goals (SDGs) transforms resource extraction into a driver of long-term regional development.
- E.g. global mining and energy companies operationalize sustainable development through renewable energy transition, circular waste processing, local enterprise funding, and formal indigenous ownership models.

**Financing clean energy transitions**

- Extractive operations provide essential raw materials—such as lithium, nickel, cobalt, copper, and rare earth elements—required to build wind turbines, solar panels, and electric vehicle batteries (SDG 7, SDG 13).
- Mining and energy operations reduce their operational footprint by powering facilities with local solar and wind microgrids and electrifying heavy haul fleets.
- E.g. BHP (Escondida, Chile): The world's largest copper mine eliminated freshwater withdrawals by transitioning 100% of its process water to desalinated seawater and sourcing all power from long-term solar and wind contracts
- E.g. Rio Tinto (Gudai-Darri, Australia): The company built a 34-megawatt solar plant integrated with a 12-megawatt-hour battery energy storage system to directly power its iron ore operations, reducing diesel reliance

**Strengthening local infrastructure and economies**

- Investments in roads, power grids, ports, and water treatment systems built for industrial projects expand public infrastructure access for surrounding rural communities (SDG 9).
- Local supply chain development: Prioritizing local procurement, vocational training, and workforce hiring creates resilient secondary business ecosystems that persist beyond the lifecycle of a resource project (SDG 8).
- E.g. Anglo American (Zimele Program, South Africa): This enterprise development initiative functions as a corporate venture fund, offering capital and business training to over 3,000 local, non-mining small businesses to foster long-term economic independence (SDG 8).

**Environmental stewardship and circularity**

- Modern operations restore ecosystems continuously throughout the extraction cycle rather than delaying biodiversity recovery and reforestation until mine decommissioning (SDG 15).
- Reprocessing historic mine waste and slag recovers valuable residual minerals while mitigating environmental contamination and reducing raw material demand (SDG 12).
- E.g. Rio Tinto (Fer et Titane, Canada): The company developed a process to extract high-purity scandium oxide—a critical mineral for fuel cells and lightweight alloys—from industrial waste tailings generated during titanium dioxide production

**Responsible governance and inclusive partnerships**

- Adhering to standards like the Extractive Industries Transparency Initiative (EITI) prevents corruption and ensures tax and royalty revenues effectively fund public health and education (SDG 16).
- Structuring formal benefit-sharing agreements and respecting Free, Prior, and Informed Consent (FPIC) empowers indigenous and local populations in land management decisions (SDG 10).
- E.g. Cameco (Northern Saskatchewan, Canada): Operating on Treaty 10 territory, the uranium miner awards over 70% of its site services to local First Nations-owned businesses and operates under legally binding Collaboration Agreements that grant communities environmental oversight and direct financial returns

<u>Some counter arguments</u>

- Extractive industries fail to contribute to sustainable development when operational practices cause permanent ecological destruction, trigger human rights abuses, or entrench economic mismanagement through the "resource curse".
- Extractive industries undermine sustainable development when operations prioritize short-term profit over long-term stability, generating externalized costs that outweigh economic returns.

**Economic vitality**

- High-volume resource exports inflate local currencies, suppressing non-resource sectors like manufacturing and agriculture. E.g. In Equatorial Guinea, oil revenues drove foreign exchange values up, rendering native agriculture non-competitive and forcing the nation to import basic food supplies.
- The majority of financial profits often flow to international parent companies or national elites rather than circulating locally. E.g. In many Sub-Saharan African mining jurisdictions, tax incentives and transfer pricing allow multinational operators to extract billions in mineral value while host governments collect minimal corporate income tax.
- Hyper-dependence on global commodity cycles leaves regional economies vulnerable to sudden market crashes. When mines unexpectedly close due to price drops, local tax bases collapse,

municipal services stall, and municipalities are left with stranded infrastructure and mass unemployment.

**Social inclusion**

- Open-pit operations frequently dispossess rural and Indigenous communities of land. E.g. The expansion of the Cerrejón coal mine in Colombia resulted in the forced displacement of several Indigenous Wayuu and Afro-Colombian communities, severing their historical ties to ancestral land and destroying subsistence farming systems.
- Unregulated supply chains exacerbate inequality and child labor. E.g. In the Democratic Republic of Congo's artisanal cobalt sector, over 40,000 children work in hazardous underground shafts without safety equipment or fair compensation to supply global battery supply chains.
- Wealthy resource concessions can fuel state repression and armed conflict. E.g. The Panguna copper mine on Bougainville Island triggered a decade-long civil war after local landowners revolted against the central government and foreign operators over severe environmental destruction and unequal distribution of revenues.

**Environmental Integrity**

- Disposal of untreated mine waste into natural waterways eliminates aquatic biomes and regional drinking sources. E.g. The Ok Tedi mine in Papua New Guinea dumped over 80 million tons of mine tailings directly into the Fly River system annually for decades, poisoning 1,000 square kilometers of forest and decimating downstream fish populations.
- Surface stripping removes topsoil, destroys ancient forests, and creates long-term carbon liabilities. E.g. The Athabasca Oil Sands in Alberta, Canada, require clearing vast tracts of boreal forest and peatlands—critical natural carbon sinks—while generating massive toxic tailings ponds that pose persistent risks to migratory birds and regional groundwater.
- Neglected structural waste systems cause acute ecocide. E.g. The 2015 Fundão tailings dam collapse in Mariana, Brazil, released 43 million cubic meters of iron ore waste into the Doce River, killing 19 people, traveling over 600 kilometers to the Atlantic Ocean, and destroying aquatic habitats along the entire river corridor.

**4** 'Transnational corporations (TNCs) are the primary drivers of interconnectedness in the global economy, rendering the regulatory power of states increasingly irrelevant.'

Evaluate this statement. **[20]**

Levels marked using the H2 generic levels descriptors for essays.

This question focuses on how Global Production Networks (GPNs) connect and create interdependence between places in the global economy. It also requires an evaluation of the relative influence of actors, comparing the coordinated power of TNCs with the role of states as regulators of economic activities. “Interconnectedness” in the global economy could be explored via international trade, global supply chains, financial flows, technological and knowledge flows.

Possible approach
Students should assess the extent to which TNCs operate autonomously across national boundaries versus how states still exert influence through varying degrees of regulation and policy to shape TNC operations within their territories. They may also consider the role of multilateral institutions (like the WTO or ASEAN) in mediating this power dynamic.

Some key points:

- Transnational corporations (TNCs) act as the primary drivers of global economic interconnectedness by orchestrating cross-border supply chains, directing foreign direct investment (FDI), expanding intra-firm trade, and standardizing global digital and physical infrastructure.

- Complex GPNs/ value chains: TNCs unbundle manufacturing processes across multiple nations to leverage local comparative advantages.

- Apple designs products in the United States, sources components from over 40 countries (e.g., displays from Samsung in South Korea, camera modules from Sony in Japan, microchips from TSMC in Taiwan), and relies on Foxconn for final assembly in China and India.
- Important influence on labour and capital flows

- Foreign direct investment and capital flows
  - TNCs inject capital, machinery, and management expertise directly into host nations, linking local supplier networks into global markets. TSMC's multi-billion-dollar semiconductor fabrication facilities built in the United States, Japan, and Germany anchor regional industrial ecosystems directly to East Asian technology and investment flows.
  - Influences labour flows too

- Intra-firm cross-border trade
  - A significant share of global trade consists of intra-firm trade—the movement of intermediate goods, services, and intellectual property between different subsidiaries of the same corporate entity.
  - Automotive manufacturers like Toyota route engines, chassis, and electronics among specialized plants in Japan, Thailand, Western Europe, and North America, binding national trade balances directly to internal corporate production schedules.

Some counter-arguments:
- Emerging government policies—such as tariffs, trade restrictions, or nearshoring mandates—are forcing TNCs to restructure these interconnected supply networks
- There are other key actors in the global economy e.g. multilateral institutions, labour
- Government policies, multilateral institutions, and labor dynamics are forcing transnational corporations (TNCs) to pivot from cost-minimized global supply chains to regionalized, high-compliance networks.
  - Tariffs, export controls, and targeted subsidies—such as the US CHIPS and Science Act and the EU Net Zero Industry Act—compel TNCs to diversify production away from single-source countries. Apple and Foxconn executed a "China+1" strategy by shifting significant electronics manufacturing to India and Vietnam, while TSMC built semiconductor plants in Arizona, Germany, and Japan.
  - Strict trade barriers force TNCs to nearshore operations to maintain market access. Automotive manufacturers like Tesla and BMW expanded parts manufacturing and assembly in Mexico to comply with the USMCA's 75% regional value content requirement.

- As the World Trade Organization's (WTO) dispute enforcement faces gridlock, TNCs must navigate overlapping mega-regional agreements like the CPTPP and RCEP, which dictate distinct regional rules for tariffs and intellectual property.
  - The OECD/G20 Global Minimum Tax framework (Pillar Two) enforces a 15% minimum effective corporate tax rate worldwide, curbing TNC profit-shifting to low-tax jurisdictions and forcing companies to make foreign direct investment decisions based on infrastructure rather than tax arbitrage.

- Labor dynamics as a driving force
  - Regional trade pacts increasingly use labor enforcement tools to police TNC supply networks. The USMCA's Rapid Response Labor Mechanism allows sanctions against specific facilities that violate union rights, prompting TNCs to enforce strict labor compliance on suppliers in key manufacturing nodes.
  - Rising labor costs in traditional manufacturing hubs like coastal China lead TNCs to migrate labor-intensive assembly to lower-cost labor markets like Vietnam and Indonesia, or to reshore operations back to high-cost regions by heavily integrating industrial automation.

**5** 'The success of fluvial flood management strategies is highly dependent on the causes of fluvial floods in the humid tropics.'

Evaluate this statement. **[20]**

Candidates can approach this question by addressing the nature of causes of fluvial flooding and how some causes lead to very high magnitude events which some flood management strategies (especially hard engineering strategies) may not be planned and designed to effectively manage. Such causes include the immense amount of high intensity rainfall brought by tropical cyclones and the consistent and relatively

heavier rain brought by the monsoons. Such causes can bring rain that overwhelms channels and cause flooding in the adjacent areas. In some cases, the magnitude of the rainfall event may be sufficiently high enough to cause hard engineering structures such as levees and dams to fail and cause floods. Specific instances of fluvial flooding such as that caused by Hurricane Katrina 2005 can be used to reflect how very high magnitude events will cause any management strategy to fail.

Candidates should evaluate the importance of other factors in contributing to the occurrence of fluvial flooding as well, considering the spatial and temporal scale of the influence.

- Planning and design of the strategy that effectively addresses the nature of floods
- Good governance and political will to provide the resources to carry out the strategies
- Level of development and access to technology
- Involvement of different stakeholders
- Geographical location and the influence of climate

**6** Evaluate the extent to which karst landform development in the humid tropics is determined by rock type. **[20]**

Possible content:
Candidates can approach this question by evaluating the importance of rock type such as limestone in the formation of key karst landforms such as sinkholes, cockpit, tower karst and caves. The significance of aspects of rocks such as chemical composition, rock purity, porosity, thickness of rock mass and joint patterns should be explored in relation to the work of geomorphic processes. The response should highlight how karst landforms and features are shaped through weathering and erosion and the role that rock characteristics play to influence the nature of karst landforms. The significance of other factors such as climate and time should also be evaluated.

Possible evaluation:
Overall, rock type is a highly significant determinant of karst landform development in the humid tropics because the presence of sufficiently soluble carbonate bedrock establishes the fundamental conditions for solution. However, it is better viewed as a prerequisite than as the sole determinant. The characteristics of the rock, particularly its purity, thickness and degree of jointing, interact with the humid tropical climate, groundwater drainage, relief and long periods of weathering to determine the resulting landforms. Thus, rock type determines the *potential* for karst development to a large extent, but the *morphology, scale and degree* of karst development are determined by the interaction of several factors.